

Module 5: Statistical Mechanics

2025

Group - A (Very Short Answer Type Questions)

1. MB statistics is applicable for photons. State True or False.

- **Answer: False**
- **Explanation:** Maxwell-Boltzmann (MB) statistics applies to **distinguishable** classical particles (like gas molecules). Photons are **indistinguishable** particles with integral spin and follow **Bose-Einstein** statistics.

2. Which statistical distribution is followed by free electrons in a conductor?

- **Answer: Fermi-Dirac Statistics**
- **Explanation:** Free electrons are **fermions** (particles with half-integer spin, $s = 1/2$) and obey the Pauli Exclusion Principle. Therefore, their distribution is governed by Fermi-Dirac statistics.

Group - C (Long Answer Type Questions)

3. Plot the Fermi distribution of a particle at $T = 0$ K and $T > 0$ K.

Solution:

The Fermi-Dirac distribution function $f(E)$ gives the probability that an energy state E is occupied by an electron:

$$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$$

Where E_F is the Fermi Energy.

- **Case 1: At Absolute Zero ($T = 0$ K)**
 - If $E < E_F$: The term $(E - E_F)$ is negative. As $T \rightarrow 0$, the exponent $\rightarrow -\infty$. Thus, $e^{-\infty} = 0$, and $f(E) = 1$.
 - If $E > E_F$: The term $(E - E_F)$ is positive. As $T \rightarrow 0$, the exponent $\rightarrow \infty$. Thus, $e^{\infty} = \infty$, and $f(E) = 0$.

- **Graph Shape:** It is a perfect **step function**. All states below E_F are filled (probability 1), and all states above E_F are empty (probability 0).
- **Case 2: At Finite Temperature ($T > 0$ K)**
 - Thermal energy excites some electrons from below E_F to above E_F .
 - At exactly $E = E_F$, the exponent is 0, so $f(E) = \frac{1}{1+1} = 0.5$.
 - **Graph Shape:** The step smoothens out. The curve deviates from the step function near E_F , showing a symmetrical "tail" where probability decreases gradually from 1 to 0.

Plot Description:

- **X-axis:** Energy (E)
- **Y-axis:** Probability $f(E)$ (Range 0 to 1)
- The $T = 0$ K line is a square edge dropping at E_F .
- The $T > 0$ K line curves smoothly through the point $(E_F, 0.5)$.

4. Compare Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac Statistics on the basis of minimum three characteristics.

Solution:

Characteristic	Maxwell-Boltzmann (MB)	Bose-Einstein (BE)	Fermi-Dirac (FD)
Type of Particle	Classical, Distinguishable	Quantum, Indistinguishable	Quantum, Indistinguishable
Spin	Any spin (Classical concept)	Integral spin ($1, 1/2, 2, \dots$) (Bosons)	Half-integral spin ($1/2, 3/2, \dots$) (Fermions)
Pauli Exclusion Principle	Does not obey	Does not obey	Obeys (Only one particle per state)
Filling of States	Any number of particles per state	Any number of particles per state	Restricted to 0 or 1 particle per state
Examples	Molecules of an ideal gas	Photons, Phonons, ^{4}He atoms	Electrons, Protons, Neutrons

2024

Group - A (Very Short Answer Type Questions)

5. What is the dimension of phase space?

- **Answer:** $2N$ (where N is the number of degrees of freedom).
- **Explanation:** Phase space is a multidimensional space where each possible state of a physical system is represented by a point. For a system with N degrees of freedom, the phase space has $2N$ dimensions (N position coordinates and N momentum coordinates). For a single particle in 3D space ($N=3$), the phase space is 6-dimensional.
- 6. At absolute zero temperature what is the probability of occupancy of an electron in an energy state above the Fermi level?
- **Answer: Zero (0)**
- **Explanation:** According to Fermi-Dirac statistics, at absolute zero (0 K), all energy states up to the Fermi energy (E_F) are completely filled (probability = 1), and all states above E_F are completely empty (probability = 0).

7. What is the average energy of electrons in a metal at 0 Kelvin?

- **Answer:** $\frac{3}{5} E_F$
- **Explanation:** At 0 K , the average energy ($\bar{E}$) of a free electron gas obeying Fermi-Dirac statistics is given by $\bar{E} = \frac{3}{5} E_F$, where E_F is the Fermi energy.

Group - B (Short Answer Type Questions)

8. Mention the criteria for the applicability of Bose-Einstein statistics.

- **Answer:**
Bose-Einstein (BE) statistics are applicable under the following criteria:
 1. **Indistinguishable Particles:** The particles must be identical and indistinguishable from one another.
 2. **Integral Spin:** The particles must have an integral spin ($0, \hbar, 2\hbar, \dots$). These particles are called **Bosons** (e.g., photons, phonons, ^{87}Rb atoms).
 3. **Symmetrical Wave Function:** The wave function of the system must be symmetric under the exchange of any pair of particles.
 4. **No Pauli Exclusion:** There is no restriction on the number of particles that can

occupy a single quantum state (unlike Fermions).

9. What do you mean by Macrostate and Microstate? Explain with suitable example.

- **Answer:**

- **Macrostate:** A macrostate describes the system using macroscopic measurable properties (like pressure, temperature, or the total number of particles in specific energy levels) without specifying the state of each individual particle.
- **Microstate:** A microstate is a specific, detailed configuration of the system that specifies the exact state (position and momentum or quantum state) of every individual particle.

- **Example:**

Consider distributing **2 distinguishable particles** (Particle A and Particle B) into **2 compartments** (Box 1 and Box 2).

- **Macrostates:** We can define the macrostate by the number of particles in each box (N_1, N_2) .
 1. Macrostate 1: (2, 0) — Both in Box 1.
 2. Macrostate 2: (1, 1) — One in Box 1, one in Box 2.
 3. Macrostate 3: (0, 2) — Both in Box 2.
- **Microstates:** We look at where specific particles (A and B) are:
 1. For Macrostate (2, 0): Only **1 Microstate** (A & B in Box 1).
 2. For Macrostate (1, 1): **2 Microstates** (A in 1, B in 2 **OR** B in 1, A in 2).
 3. For Macrostate (0, 2): Only **1 Microstate** (A & B in Box 2).
- **Conclusion:** The macrostate (1, 1) is the most probable because it corresponds to the highest number of microstates (2).

2023

Group - A (Very Short Answer Type Questions)

10. Fill in the blank: Planck's formula for black body radiation can be derived from _____ statistics.

- **Answer:** Bose-Einstein
- **Explanation:** Planck's law describes the spectral density of electromagnetic radiation emitted by a black body. Since radiation consists of **photons** (which are indistinguishable particles with integral spin, i.e., Bosons), their statistical behavior is governed by **Bose-Einstein statistics**.

11. In which statistics, Pauli's exclusion principle is valid?

- **Answer: Fermi-Dirac Statistics**
- **Explanation:** Pauli's Exclusion Principle states that no two identical fermions can occupy the same quantum state simultaneously. Fermi-Dirac statistics applies specifically to **Fermions** (particles with half-integer spin like electrons, protons, neutrons) and inherently incorporates this principle.

12. At absolute zero temperature, what is the Fermi distribution of electrons when the energy of the system is greater than Fermi energy?

- **Answer: Zero (0)**
- **Explanation:** The Fermi-Dirac distribution function is:

$$f(E) = \frac{1}{e^{(E-E_F)/k_B T} + 1}$$

At Absolute Zero ($T = 0$ K):

- If Energy (E) > Fermi Energy (E_F), the exponent becomes $+\infty$ (since we are dividing a positive number by 0).
- $\frac{1}{\infty} \rightarrow 0$
- Therefore, $f(E) = \frac{1}{\infty} = 0$.
- **Physical Meaning:** At absolute zero, it is impossible to find an electron with energy higher than the Fermi level; all such states are empty.

Distribution (Energies)	Microstates Calculation	Number of Microstates
{1E, 1E, 4E}	$\frac{3!}{2!} = 3$	3
{1E, 2E, 3E}	$3! = 6$	6

{2E, 2E, 2E}	$3!/3$	1
Total		10

2017

Group - A (Multiple Choice Type Questions)

18. Energy emitted from the surface of a black body at an absolute temperature T is proportional to

- Answer: (d) T^4
- Explanation: According to **Stefan-Boltzmann Law**, the total energy radiated per unit surface area of a black body per unit time is directly proportional to the fourth power of the black body's thermodynamic temperature.

$$E = \sigma T^4$$

(Where σ is the Stefan-Boltzmann constant).

19. The de Broglie wavelength of a thermal neutron is given by

- Answer: (c) $\frac{h}{\sqrt{3mkT}}$
- Explanation:
 - Neutrons in thermal equilibrium at temperature T behave like a gas obeying **Maxwell-Boltzmann statistics**.

- The average kinetic energy of a gas particle (neutron) is $K = \frac{3}{2}kT$.

- Momentum $p = \sqrt{2mK} = \sqrt{2m \left(\frac{3}{2}kT \right)} = \sqrt{3mkT}$.

- de Broglie wavelength $\lambda = \frac{h}{p} = \frac{h}{\sqrt{3mkT}}$.

Group - C (Long Answer Type Questions)

20. Write down the Planck's law of frequency distribution of black body radiation.

• **Answer:**

Planck's law describes the spectral density of electromagnetic radiation emitted by a black body in thermal equilibrium at a given temperature T .

The energy density $u(\nu)d\nu$ inside a cavity in the frequency range ν to $\nu + d\nu$ is given by:

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu$$

- h = Planck's constant
- c = Speed of light
- k_B = Boltzmann constant
- ν = Frequency of radiation

21. Show graphically how the energy density vs frequency plot of black body radiation is changed if the temperature is changed.

• **Answer:**

* **Graph Description:**

* **X-axis:** Frequency (ν) or Wavelength (λ).

* **Y-axis:** Energy Density ($u(\nu)$) or Intensity.

* **Curves:** Multiple curves are drawn for different temperatures ($T_1 < T_2 < T_3$).

○ **Key Features:**

1. **Higher Peak:** As temperature increases (T), the total energy radiated (area under the curve) increases significantly.
2. **Peak Shift:** The peak of the curve shifts towards higher frequencies (or shorter wavelengths) as temperature increases (Wien's Displacement Law).
3. **Non-intersecting:** The curve for a higher temperature lies entirely above the curve for a lower temperature; it emits more energy at every frequency.

22. Derive Wien's displacement law from Planck's law of Black body radiation.

• **Answer:**

Step 1: Start with Planck's Law in terms of wavelength (λ):

$$u(\lambda)d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1} d\lambda$$

Step 2: Find the maximum intensity (I_{max}):

To find the peak, we differentiate $u(\lambda)$ with respect to λ and set it to zero:

$$\frac{du(\lambda)}{d\lambda} = 0$$

Step 3: Substitution:

Let $x = \frac{hc}{\lambda k_B T}$. Thus $\lambda = \frac{hc}{x k_B T}$.

Substituting this into the derivative and solving the resulting transcendental equation (usually solved numerically) gives a constant value for x :

$$x \approx 4.965$$

Step 4: Solve for λ_{max} :

$$\frac{hc}{\lambda_{max} k_B T} = 4.965$$

$$\lambda_{max} T = \frac{hc}{4.965 k_B}$$

Step 5: Conclusion:

The Right Hand Side consists of constants (h, c, k_B), which combine to form **Wien's constant** ($b \approx 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$).

$$\lambda_{max} T = b = \text{Constant}$$

This is Wien's Displacement Law.

2016

Group - A (Multiple Choice Type Questions)

23. According to Wien's displacement law

- **Answer:** (D) $\lambda_m T = \text{Constant}$
- **Explanation:**

Wien's Displacement Law describes the relationship between the temperature of a black body and the wavelength at which it emits the most radiant energy.

It states that the wavelength corresponding to maximum intensity (λ_m) is inversely proportional to the absolute temperature (T).

$$\lambda_m \propto \frac{1}{T} \implies \lambda_m T = b$$

(Where b is Wien's displacement constant, approximately $2.898 \times 10^{-3} \text{ m} \cdot \text{K}$).

2015

Group - A (Multiple Choice Type Questions)

24. According to Rayleigh-Jean's Law, the energy density in a blackbody radiation u_λ is

- **Answer:** (A) inversely proportional to the fourth power of wavelength

- **Explanation:**

Rayleigh-Jeans law, derived using classical statistics, states that the energy density u_λ is given by:

$$u_\lambda d\lambda = \frac{8\pi k_B T}{\lambda^4} d\lambda$$

Therefore, $u_\lambda \propto \frac{1}{\lambda^4}$.

25. de Broglie wavelength of a gas molecule at temperature T is

- **Answer:** (C) $\frac{h}{\sqrt{3mkT}}$
- **Explanation:**
 - Gas molecules follow **Maxwell-Boltzmann statistics**. The average kinetic energy (K) of a gas molecule at temperature T is given by $K = \frac{3}{2}kT$.
 - Momentum $p = \sqrt{2mK} = \sqrt{2m \left(\frac{3}{2}kT \right)} = \sqrt{3mkT}$.
 - The de Broglie wavelength is $\lambda = \frac{h}{p}$.
 - Substituting p , we get $\lambda = \frac{h}{\sqrt{3mkT}}$.

Group - B (Short Answer Type Questions)

26. What is ultraviolet catastrophe?

- **Answer:**

The **Ultraviolet Catastrophe** was a major failure of classical physics (specifically the Rayleigh-Jeans law) to explain black body radiation.

 - **Classical Prediction:** The Rayleigh-Jeans law predicted that the energy density of radiation emitted by a black body was proportional to ν^2 (or $1/\lambda^3$). This meant that as frequency increased (into the Ultraviolet range) or wavelength decreased, the emitted energy should increase towards infinity.
 - **Experimental Reality:** Experiments showed that intensity peaks at a certain wavelength and then drops to zero at very short wavelengths (high

frequencies).

- **Significance:** This discrepancy demonstrated that classical mechanics could not explain thermal radiation, leading Max Planck to introduce the concept of **quantization of energy** (Bose-Einstein statistics precursor), which resolved the catastrophe.

Group - C (Long Answer Type Questions)

27. (a) What do you mean by black body? State Kirchhoff's law of blackbody radiations.

- **Answer:**

- **Black Body:** An ideal physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence. Because it reflects no light, it appears perfectly black at low temperatures. A black body is also a perfect emitter of radiation at every wavelength for a given temperature.
- **Kirchhoff's Law:** It states that at a given temperature (T), the ratio of the **emissive power** (E_λ) to the **absorptive power** (a_λ) for a given wavelength is constant for all bodies and is equal to the emissive power of a perfectly black body (E_λ^0) at that same temperature.

$$\frac{E_\lambda}{a_\lambda} = E_\lambda^0$$

This implies that good absorbers are also good emitters.

28. (b) Establish Newton's law of cooling from the Stefan-Boltzmann law of Blackbody radiation.

- **Answer:**

Step 1: Stefan-Boltzmann Law

The rate of heat loss (P) per unit area for a body at temperature T in surroundings at temperature T_0 is:

$$P = \varepsilon \sigma (T^4 - T_0^4)$$

(Where ε is emissivity and σ is Stefan's constant).

Step 2: Small Temperature Difference Approximation

Assume the temperature difference is very small: $T = T_0 + \Delta T$, where $\Delta T \ll T_0$.

Substitute T into the equation:

$$P = \varepsilon \sigma ((T_0 + \Delta T)^4 - T_0^4)$$

Step 3: Binomial Expansion

Using the expansion $(1 + x)^n \approx 1 + nx$ for $x \ll 1$:

$$(T_0 + \Delta T)^4 = T_0^4 \left(1 + \frac{\Delta T}{T_0}\right)^4 \approx T_0^4 \left(1 + 4\frac{\Delta T}{T_0}\right)$$

$$(T_0 + \Delta T)^4 \approx T_0^4 + 4T_0^3 \Delta T$$

Step 4: Substitution

$$P = \varepsilon \sigma (T_0^4 + 4T_0^3 \Delta T - T_0^4)$$

$$P = \varepsilon \sigma (4T_0^3 \Delta T)$$

Step 5: Conclusion

Since ε, σ, T_0 are constants for a given surrounding, let $k = 4\varepsilon \sigma T_0^3$.

$$P = k \Delta T \implies \frac{dQ}{dt} \propto (T - T_0)$$

This matches **Newton's Law of Cooling**: The rate of cooling is directly proportional to the temperature difference between the body and its surroundings.

2012

Group - A (Multiple Choice Type Questions)

29. The emissive power of a black body kept at absolute temperature T which is very near to the temperature of surroundings (T_0) is proportional to

- **Answer:** (b) $T - T_0$
- **Explanation:** According to Stefan-Boltzmann Law, the net energy radiated is $E \propto (T^4 - T_0^4)$. If $T \approx T_0$, let $T = T_0 + \Delta T$ where ΔT is small. Using the binomial expansion: $E \propto ((T_0 + \Delta T)^4 - T_0^4) \approx (T_0^4 + 4T_0^3 \Delta T - T_0^4) \propto 4T_0^3 \Delta T$. Since T_0 is constant, $E \propto \Delta T$, i.e., $E \propto (T - T_0)$. This leads to **Newton's Law of Cooling**.

30. According to Wien's displacement law

- **Answer:** (a) $\lambda_m T = \text{Constant}$
- **Explanation:** Wien's displacement law states that the product of the wavelength corresponding to the maximum intensity (λ_m) and the absolute temperature (T) is a constant.

Group - B (Short Answer Type Questions)

31. Define Black-Body? Establish Wein's distribution law and Stefan's law from Plank's black body radiation law.

- **Answer: 1. Definition of Black Body:** A black body is an idealized physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence. It reflects nothing and transmits nothing, appearing perfectly black at cold

temperatures. It is also a perfect emitter, radiating the maximum possible energy for a given temperature.

- **2. Establishing Stefan's Law from Planck's Law:**

- **Planck's Law (Energy Density):**

$$u(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} d\nu$$

- **Total Energy Density (U):**

Integrate over all frequencies (0 to ∞):

$$U = \int_0^\infty \frac{8\pi h\nu^3}{c^3(e^{h\nu/kT} - 1)} d\nu$$

- **Substitution:** Let $x = \frac{h\nu}{kT} \Rightarrow \nu = \frac{xkT}{h}$ and $d\nu = \frac{kT}{h} dx$.

$$U = \frac{8\pi h}{c^3} \left(\frac{kT}{h}\right)^4 \int_0^\infty \frac{x^3}{e^x - 1} dx$$

- **Integration:** The standard integral

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$$

- **Result:**

$$U = \frac{8\pi^5 k^4}{15c^3 h^3} T^4$$

$$U = aT^4$$

- The total emissive power E is related to energy density by $E = \frac{c}{4} U$.

$$E = \sigma T^4$$

(Where σ is Stefan's Constant).

- **3. Establishing Wien's Law from Planck's Law:**

(Note: "Wien's distribution law" can refer to the high-frequency limit or the displacement law. Deriving the Displacement Law is the standard requirement).

- **Planck's Law (Wavelength form):**

$$u(\lambda) = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

- **Maximize Intensity:** To find the peak wavelength λ_m , differentiate $u(\lambda)$ with respect to λ and set to 0.

$$\frac{du}{d\lambda} = 0$$

- **Solving:** Let $x = \frac{hc}{\lambda kT}$. The differentiation leads to the transcendental equation:

$$\frac{xe^x}{x-1} - 5 = 0$$

- **Solution:** The root is $x \approx 4.965$.

- **Result:**

$$\frac{hc}{\lambda_m kT} = 4.965$$

$$\lambda_m T = \frac{hc}{4.965k} = \text{Constant}(b)$$

This is **Wien's Displacement Law**.

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